



DATA MINING ANALYSIS REPORT

HOUSE PRICING ANALYSIS

Date: 31.12.2025

Data Mining Techniques with Python

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Introduction:

This report shows a data mining analysis from *House Prices* dataset downloaded from Kaggle. The main purpose of the dataset is to identify key factors that have an impact on house prices. Because of that, we aim to transform raw data into meaningful insights. That is why I applied analytical concepts such as data cleaning, transformation, visualization, and interpretation using tools such as Jupiter Notebook, Python and visualization, matplotlib, and seaborn.

Techniques Used:

- Exploratory Data Analysis (EDA)
- Data Preprocessing and Cleaning
- Data Visualization
- Regression Algorithms

DATASET OVERVIEW

Kaggle: House Prices – Advanced Regression Techniques

File: train.csv

Data Frame Shape is: (1460, 81)

	Id	MSSubClass	MSZoning	LotFrontage	LotArea	Street	Alley	LotShape	LandContour	Utilities	...	PoolArea	PoolQC	Fence	MiscFeature	MiscVal	MoS
0	1	60	RL	65.0	8450	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	NaN	NaN	0	
1	2	20	RL	80.0	9600	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	NaN	NaN	0	
2	3	60	RL	68.0	11250	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	NaN	NaN	0	
3	4	70	RL	60.0	9550	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	NaN	NaN	0	
4	5	60	RL	84.0	14260	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	NaN	NaN	0	

5 rows × 81 columns

Feature Name	Description	Data Type
SalePrice	Sale price of the house	Numerical
GrLivArea	Above ground living area (sq ft)	Numerical
OverallQual	Overall material and finish quality	Object
GarageCars	Size of garage in car capacity	Numerical
TotalBsmtSF	Total basement area	Numerical

DATA PREPROCESSING:

Data preprocessing phase for that we find the missing values and their percentage.

Missing Value Analysis:

PoolQC 1453	
<i>MiscFeature</i>	1406
<i>Alley</i>	1369
<i>Fence</i>	1179
<i>FireplaceQu</i>	690
<i>LotFrontage</i>	259
<i>GarageYrBlt</i>	81
<i>GarageCond</i>	81
<i>GarageType</i>	81
<i>GarageFinish</i>	81
<i>GarageQual</i>	81
<i>BsmtFinType2</i>	38
<i>BsmtExposure</i>	38
<i>BsmtQual</i>	37
<i>BsmtCond</i>	37
<i>BsmtFinType1</i>	37
<i>MasVnrArea</i>	8
<i>MasVnrType</i>	8
<i>Electrical</i>	1
<i>Id</i>	0

Null Values Percentage (%)	
PoolQC	99.52%
MiscFeature	96.30%
Alley	93.77%
Fence	80.75%
FireplaceQu	47.26%
LotFrontage	17.74%
GarageYrBlt	5.55%
GarageCond	5.55%
GarageType	5.55%
GarageFinish	5.55%

Preprocessing Steps Applied

if missing values over than %50 percent we dropped them from dataset. For other null values, the percentage was lower than %50, so I applied the numerical values median method to fill missing values and gained them to dataset. For categorical missing values filled with 'none' values. With that there is not any missing value left.

Sample Preprocessing Code

Dropped if missing value more than %50 percent

```
df = df.loc[:, df.isnull().mean() < 0.5]
```

```
lot_median = df['LotFrontage'].median()
```

```
df['LotFrontage'] = df['LotFrontage'].fillna(lot_median)
```

```
df['GarageYrBlt'] = df['GarageYrBlt'].fillna(0)
```

FireplaceQu, GarageType, GarageFinish etc. are categorical values. So we use none label.

```
df_categorical = [
```

```
    'FireplaceQu',
```

```
    'GarageType',
```

```
    'GarageFinish',
```

```
    'GarageQual',
```

```
    'GarageCond',
```

```
    'BsmtExposure',
```

```
    'BsmtFinType2',
```

```
    'MasVnrArea'
```

```
]
```

```
for col in df_categorical:
```

```
    if col in df.columns:
```

```
        df[col] = df[col].fillna("None")
```

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was applied to understand the effect of home features on sales prices and the distribution of variables.

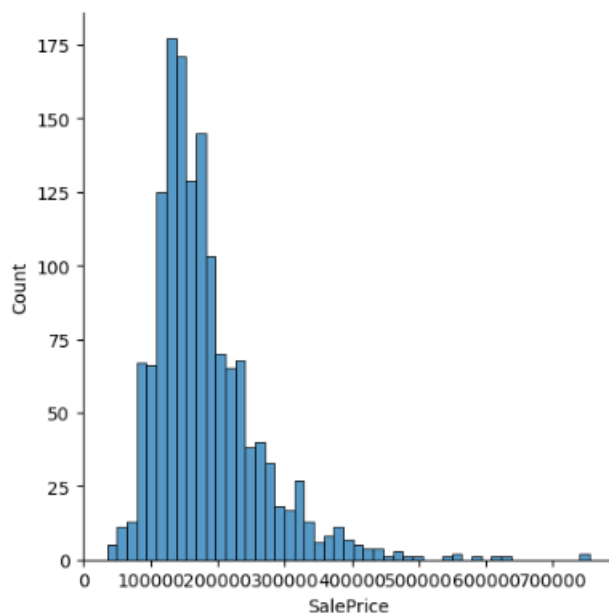
Descriptive Statistics

	SalePrice	GrLivArea	YearBuilt	OverallQual	LotArea
Mean	180921.2	1515.46	1971.27	6.10	10516.83
Median	163000.0	1464.00	1973.00	6.00	9478.50
Std Dev	79442.5	525.48	30.20	1.38	9981.26
Min	34900.0	334.00	1872.00	1.00	1300.00
Max	755000.0	5642.00	2010.00	10.00	215245.00

Visualization

Target Variable Analysis (SalePrice) :

Histogram is used for show distribution of house prices and this show us right-skewed histogram result.

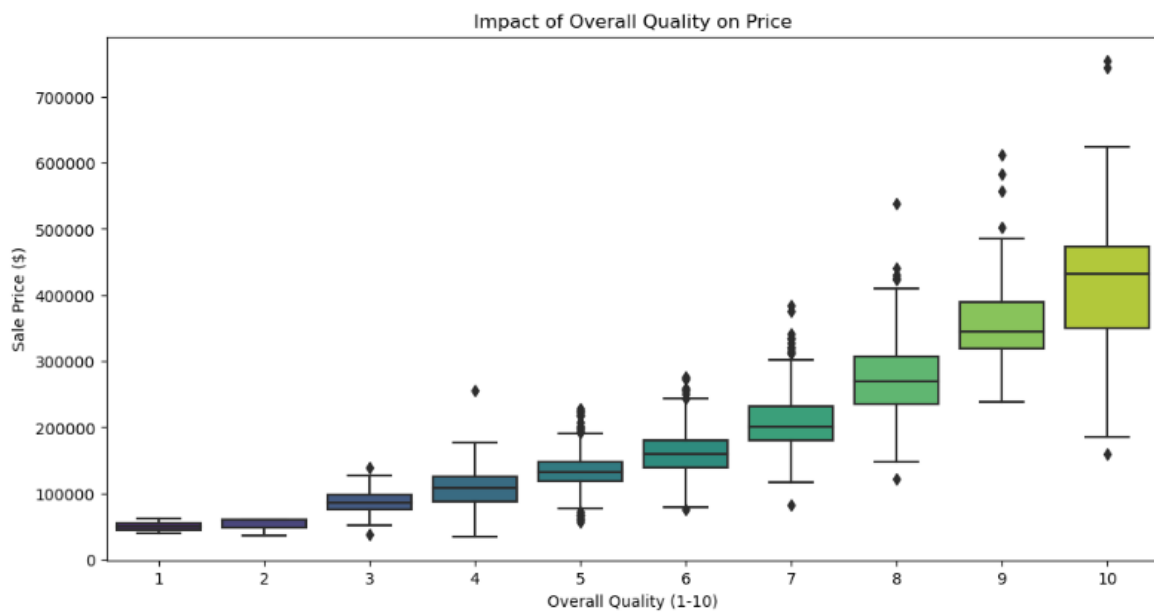


GrvLivArea Analysis:

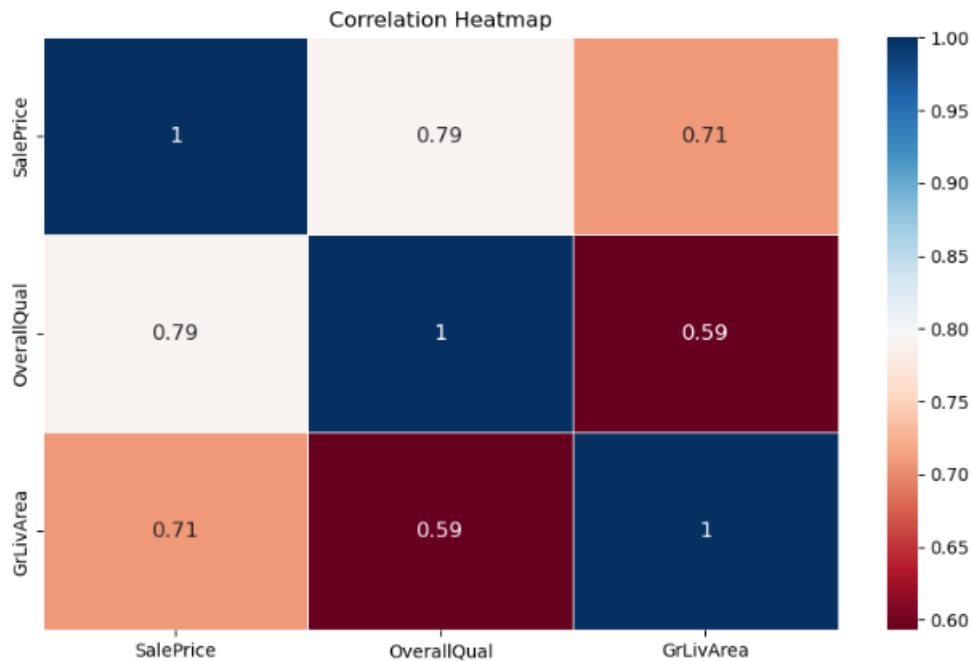
Scatter plot with best fit line used for shown GrvLivArea analysis. The visualization reveals a strong positive linear relationship



OverallQual Analysis:



Correlation HeatMap



Key Findings:

- Correlation Heatmap: Overall Quality, the correlation heat map showed us that 'Overall Quality' is the strongest predictor of home prices. As the quality score increases, so do prices.
- Scatter Plot and Best Fit Line: Result is houses with larger living areas tend to have higher sale prices.
- Overall Quality show us if house quality number is higher price is higher.
- Histogram show that most house are sold lower prices.

DATA MINING TECHNIQUES

Regression Analysis is used because predict house prices based on numerical values. The regression model achieved an R^2 score of 0.79. The RMSE value shows that the model provides reasonably accurate price predictions.

```
RMSE: 39763.29526578046
R2 Score: 0.7938653966356576
```

Features used is in regression model such as , 'OverallQual', 'GrLivArea', 'TotalBsmtSF', 'GarageCars', 'YearBuilt'.

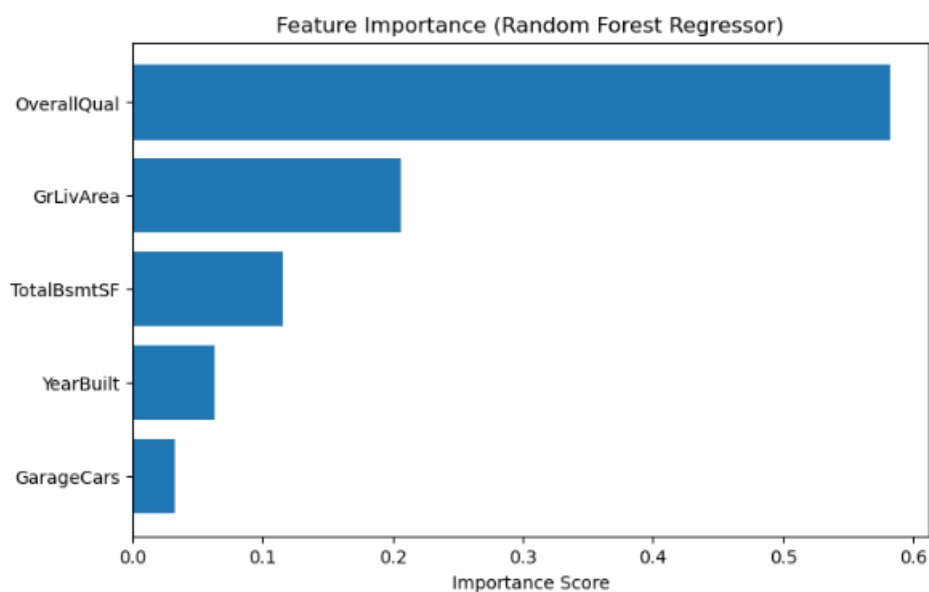
Random Forest Analysis Output:

```
Random Forest RMSE: 28893.094581024354
Random Forest R2 Score: 0.8911635708940517
```

Features used such as, 'OverallQual', 'GrLivArea', 'TotalBsmtSF', 'GarageCars', 'YearBuilt'.

Feature Importance

Random Forest analysis show us ranking of values. As a result we can say overallQual is the most important score have most impact on sale price.



Python Code Examples

Preprocessing Code

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

df = pd.read_csv("train.csv")

df.head()

df.info()

print("-----")

print("Data Frame Shape is:")

df.shape

print("Find Null or Nan values")

df.isnull().sum().sort_values(ascending=False).head(20)

percentage_of_null = df.isnull().mean()*100

shown_percentage = percentage_of_null.sort_values(ascending=False).head(10)

display(shown_percentage.to_frame(name='Null Values Percentage (%)').style.format("{:.2f}%"))

df = df.loc[:, df.isnull().mean() < 0.5]

df_column = ['SalePrice', 'GrLivArea', 'YearBuilt', 'OverallQual', 'LotArea']

df_table = df[df_column].describe()

df_table = df_table.loc[['mean', '50%', 'std', 'min', 'max']]

df_table.index = ['Mean', 'Median', 'Std Dev', 'Min', 'Max']

df_table = df_table.round(2)

print(df_table)
```

Data Visualization

Histogram:

```
plt.figure(figsize=(10,9))

sns.displot(df["SalePrice"])
```

GrvLivArea Analysis:

```
plt.figure(figsize=(10,6))

sns.regplot(
    x='GrLivArea',
    y='SalePrice',
```

Deniz İlkiz

```
data=df,
line_kws={'color':'red'}
)
plt.title("Living Area vs Sale Price with Best Fit Line")
plt.xlabel("Above Ground Living Area (sqft)")
plt.ylabel("Sale Price")
plt.show()
OverallQual Analysis:
plt.figure(figsize=(12, 6))
sns.boxplot(x='OverallQual', y='SalePrice', data=df, palette='viridis')
plt.title('Impact of Overall Quality on Price')
plt.xlabel('Overall Quality (1-10)')
plt.ylabel('Sale Price ($)')
plt.show()
Correlation Heatmap:
plt.figure(figsize=(10,6))
corr = df[['SalePrice', 'OverallQual', 'GrLivArea']].corr()
sns.heatmap(corr,
            annot=True,
            linewidths=0.5,
            annot_kws={"size": 12},
            linecolor="w",
            cmap="RdBu")
plt.title("Correlation Heatmap")
plt.show()
Regression code:
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
features = ['OverallQual', 'GrLivArea', 'TotalBsmtSF', 'GarageCars', 'YearBuilt']
X = df[features]
y = df['SalePrice']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
r2 = r2_score(y_test, y_pred)
print("RMSE:", rmse)
print("R2 Score:", r2)

Random Forest:

from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error, r2_score

features = ['OverallQual', 'GrLivArea', 'TotalBsmtSF', 'GarageCars', 'YearBuilt']
X = df[features]
y = df['SalePrice']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

rf_model = RandomForestRegressor(
    n_estimators=100,
    random_state=42
)

rf_model.fit(X_train, y_train)
y_pred_rf = rf_model.predict(X_test)
rmse_rf = np.sqrt(mean_squared_error(y_test, y_pred_rf))
r2_rf = r2_score(y_test, y_pred_rf)
print("Random Forest RMSE:", rmse_rf)
print("Random Forest R2 Score:", r2_rf)

Feature importance

importances = rf_model.feature_importances_
feature_importance_df = pd.DataFrame({
    'Feature': features,
    'Importance': importances
}).sort_values(by='Importance', ascending=True)

plt.figure(figsize=(8,5))

plt.barh(feature_importance_df['Feature'], feature_importance_df['Importance'])

plt.xlabel('Importance Score')
```

```
plt.title('Feature Importance (Random Forest Regressor)')  
plt.show()
```

Results

Key Findings

- Correlation Heatmap: Overall Quality, the correlation heat map showed us that 'Overall Quality' is the strongest predictor of home prices. As the quality score increases, so do prices.
- Scatter Plot and Best Fit Line: Result is houses with larger living areas tend to have higher sale prices.
- Overall Quality show us if house quality number is higher price is higher.
- Histogram show that most house are sold lower prices.
- Data Mining Techniques show us that The Random Forest Regressor best performance then the Linear Regression model by achieving a higher R^2 score and a lower RMSE value. Random forest R score is 89% and regression is 79% .

Conclusion:

In this project, the key factors that have an impact on house sales prices have been successfully analyzed and gained meaningful results. To gain these results, data cleaning, transformation, and visualization steps are applied. With that, I understand the importance of data prep and exploratory data analysis (EDA) in understanding the real estate market. Furthermore, data mining techniques were applied such as regression model and random forest models used for predict house prices.